EXECUTIVE USE CASE · AXA INSURANCE

Governed loss-ratio intelligence

Let an executive ask, in plain language, **which parts of the book are underperforming on loss ratio** — and get the **one governed number** behind it in seconds, not weeks. Built on the Databricks lakehouse the business already runs, so people and AI agents share the same truth.

64.6% PORTFOLIO LOSS RATIO	**91.8%** COMBINED RATIO
Seconds NOT WEEKS — SELF-SERVE	**1** DEFINITION, GOVERNED & CERTIFIED

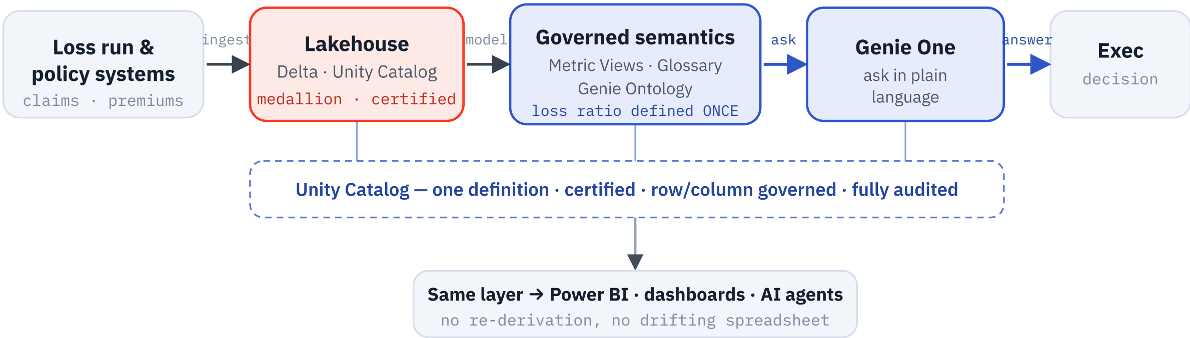
THE USE CASE

One question. One number. Trusted.

As a Chief Underwriting Officer / COO, **I want to** ask which lines of business are running hot on loss ratio this quarter and see the approved definition behind it — **so that I** can direct underwriting, pricing and reserving action with confidence.

> **Today:** the number arrives in a slide three weeks late, and "loss ratio" means something different in every deck — paid vs incurred, gross vs net, written vs earned premium. The decision waits on an analyst, and nobody is sure whose definition is on screen.

THE FLOW — DATA TO DECISION



The question routes through the governed layer. Loss ratio is defined once (incurred ÷ earned) as a metric view; Genie answers off that definition, and the very same layer feeds BI and agents — so every consumer returns the same, certified number.

HOW THE ONTOLOGY IS BUILT

01 · GOLD LAYER

Gold Layer

Certify the P&C tables — business descriptions, declared keys, trusted.

02 · METRIC VIEWS

Metric Views

Define loss ratio & combined ratio **once**, as governed code.

03 · GENIE AGENT

Genie Agent

A conversational agent on the tables + metric views.

04 · DOMAIN + GLOSSARY

Domain & Glossary

Business terms as pages; every asset tagged into the Insurance domain.

RESULT

Genie Ontology

The governed context graph — Unity Catalog semantics + auto-learned knowledge.

Built bottom-up on the lakehouse (notebooks 01→03), then **discovered top-down** at the domain (04). Metric Views + Glossary + Domain **are** the Unity Catalog semantics; combined with auto-learned knowledge they form the **Genie Ontology** — the one governed meaning that both the Genie Agent and business users reason over.

DOCUMENTATION & STEPS

- Workshop repo & README ↗
- Step 0 — Generate data `scripts/generate_synthetic_data.py` ↗
- Steps 01–04 — Notebooks ↗
- Exec story — loss ratio ↗

WHAT THE EXECUTIVE SHOULD BE LOOKING AT

1 Level, target & trend
Loss ratio vs. target and prior year — a point value means nothing without a bar and a trend.

2 Which segment drives it
Portfolio 64.6% can hide a line of business at 90%+. Act on the segment, not the average.

3 Frequency vs. severity
More claims or bigger claims? Different fixes — terms & risk selection vs. pricing & cat exposure.

4 Combined ratio & trust
Underwriting profit is loss + expense (91.8%). And is the number from a certified, current asset?

WHY IT MATTERS

Faster decisions
Self-serve in seconds — the analyst backlog and the three-week slide disappear.

One source of truth
Loss ratio defined once; every deck, dashboard and agent agrees.

Trusted & auditable
Certified assets, governed access, full lineage — safe for regulated reporting.

Scales to AI
Agents inherit the same governed meaning — no ungoverned model of the business.